

Learning to See Sharper: A Physics-Informed Artificial Intelligence Framework for Super-Resolving Galaxy Spectra

The scientific information recoverable from galaxy spectra is fundamentally limited by spectral resolution. High-resolution spectroscopy enables precise redshift measurements for cosmology and reveals key galaxy properties, including star-formation rates, gas-phase metallicity, and active galactic nucleus activity. However, many important emission lines become blended in low-resolution spectra, introducing systematic uncertainties that limit both galaxy evolution studies and cosmological analyses [6, 1, 14, 7].

Current facilities each face a trade-off between spectral resolution, survey area, and depth. Large space missions such as *Euclid* [8] and the forthcoming *Roman Space Telescope* [15, 12] observe vast numbers of distant galaxies using low-resolution spectroscopy. In contrast, the *James Webb Space Telescope* (JWST) [4] delivers exceptional spectral resolution but only over small fields of view.

My research addresses this challenge by developing deep-learning methods that reconstruct higher-resolution spectra from low-resolution observations. Using paired spectra from the *JWST Advanced Deep Extragalactic Survey* (JADES) [2], where low-resolution prism observations are matched with medium-resolution grating spectra of the same galaxies, I train neural networks to recover diagnostic spectral features that would otherwise remain unresolved [5].

I will then extend this framework to enhance the scientific return of simulated spectra expected from the future *Roman Space Telescope*, enabling more accurate galaxy property measurements and improved cosmological constraints from millions of galaxies.

Summer 2027 at Infrared Processing and Analysis Center (IPAC):

During my time at IPAC, I will focus on four components: (1) Exploring and measuring the effect of super-resolving galaxy spectra on the performance of downstream science tasks including galaxy diagnostics; (2) Comparing our developed technique against classical deconvolution methods [9, 10, 13]; (3) Incorporating the framework into the official NASA-supported spectroscopic data pipelines; (4) Adapting the framework to Roman grism with IPAC's mission team, incorporating photometry to aid redshift prediction. Timing is critical: Roman's first data release is expected Summer 2027. On-site access to mission data and Dr. Shoubaneh Hemmati (research scientist of *Euclid* and *Roman* missions) enables immediate validation impossible remotely. In-person collaboration ensures our methodology aligns with mission requirements and goals.

This work advances galaxy evolution studies in astrophysics while demonstrating how machine learning extracts maximum value from observational data [11, 3]. All products (e.g., code, trained models, and super-resolved spectra) will be publicly released with proper documentation for community use, extending the science reach of JWST, Roman, *Euclid*, and future missions.

References

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